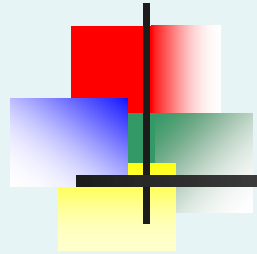
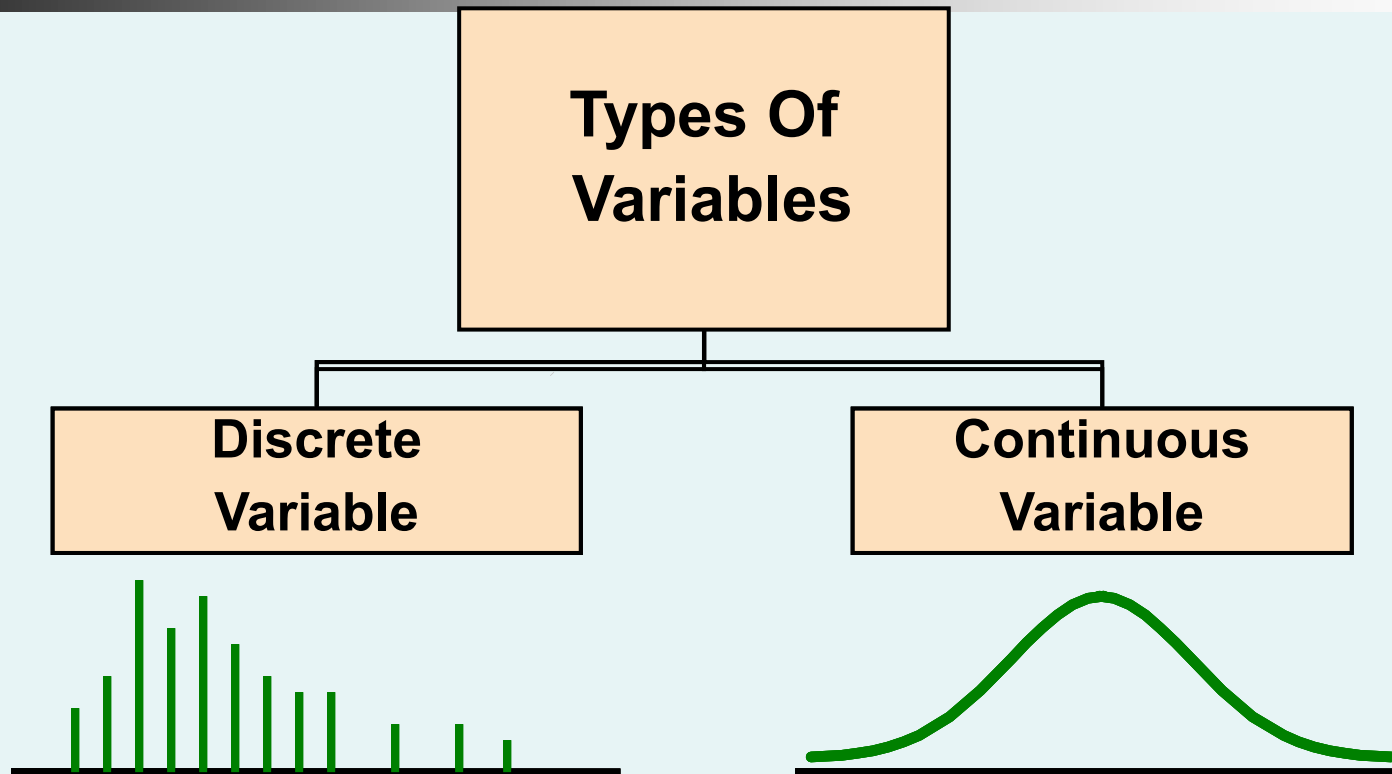


Expected Value and Variance

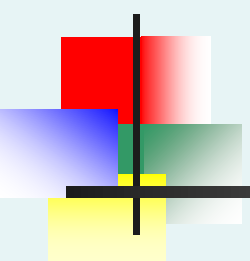


Types Of Variables



Discrete variables take only separate, isolated values.

Continuous variables can take any value in an interval.

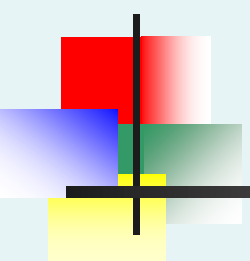


Probability Distribution For A Discrete Random Variable

- A ***probability distribution for a discrete random variable*** is a list of its all possible values along with the corresponding probabilities.

- Example:

x values of X	P(x) probabilities
0	1/8
1	3/8
2	3/8
3	1/8



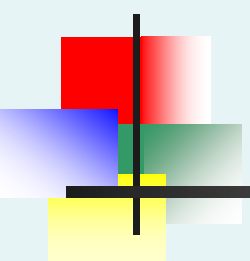
Expected Value

- **Expected Value** of a random variable is the mean, or the average:

$$\mu = E(X) = \sum_x xP(x)$$

- **Example:** Randomize 2 patients,
 $X = \#$ in the placebo group,
compute expected value of X :
 $E(X) = (0)(0.25) + (1)(0.50) + (2)(0.25)$
 $= 1.0$

x	P(x)
0	0.25
1	0.50
2	0.25

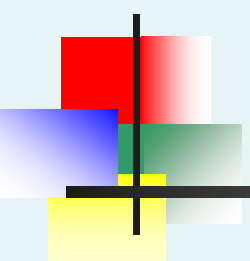


Variance and Standard Deviation

- **Variance** of a discrete random variable measures the spread, or amount of uncertainty, or how far its values can be from the mean:

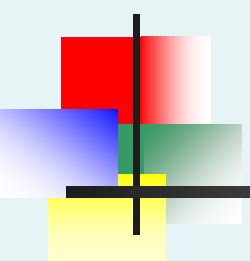
$$\sigma^2 = \text{Var}(X) = E(X - \mu)^2 = \sum_x [x - E(X)]^2 P(x)$$

- **Standard Deviation** of a random variable is the square root. $\sigma = \sqrt{\sigma^2}$ It has the same measurement unit as the original variable X .



Covariance

- The covariance measures the strength of the linear relationship between two discrete random variables X and Y .
- A positive covariance indicates a positive relationship.
- A negative covariance indicates a negative relationship.



The Covariance Formula

- The **covariance** formula:

$$\sigma_{XY} = E(X - \mu_X)(Y - \mu_Y) = \sum_x \sum_y (x - EX)(y - EY)P(x, y)$$

where: X = discrete random variable X

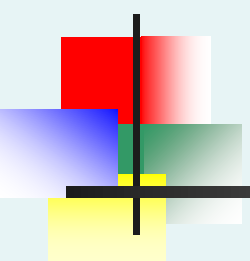
x = value of X

Y = discrete random variable Y

y = value of Y

$P(x, y) = P(X=x \text{ and } Y=y)$

= probability of a joint event $\{X=x \text{ and } Y=y\}$



The Sum of Two Random Variables

- Expected Value of the sum of two random variables:

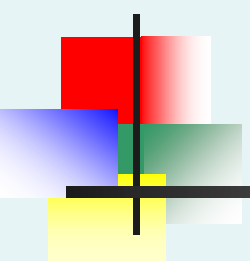
$$E(X + Y) = E(X) + E(Y)$$

- Variance of the sum of two random variables:

$$\text{Var}(X + Y) = \sigma_{X+Y}^2 = \sigma_X^2 + \sigma_Y^2 + 2\sigma_{XY}$$

- Standard deviation of the sum of two random variables:

$$\sigma_{X+Y} = \sqrt{\sigma_{X+Y}^2}$$



The Weighted Sum Two Random Variables

- Expected Value of a weighted sum:

$$E(aX + bY) = aE(X) + bE(Y)$$

- Variance of a weighted sum:

$$\text{Var}(aX + bY) = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_{XY}$$

- Standard deviation of a weighted sum:

$$\sigma_{aX+bY} = \sqrt{\sigma_{aX+BY}^2}$$